

100 Days of Azure DevOps

Day 3 Handout — Every Portal click is an ARM call — tags, locks, providers

1 - Azure & DevOps Foundations · Personal lab only · Educational content · Not a sales pitch

Architecture A — request path (you never talk to the VM first)

Layer	ARM job	Lab command / type
Client	Portal / az / PowerShell / SDK / pipeline	az group create, az lock create
ARM control plane	AuthN/Z, validate schema, orchestrate	management.azure.com REST
Management group	Policy + RBAC above subscriptions	Skip for personal lab; know it exists
Subscription	Quota, billing, RBAC assignment scope	az account show
Resource group	Deploy scope + delete scope + tag rollout	Microsoft.Resources/resourceGroups
Resource provider	Microsoft.Storage, Microsoft.Web, Microsoft.Compute	az provider list --query [].namespace
Resource	The billed object	Do not create a VM today — RG + lock only

Architecture B — decide tag vs lock vs policy

Need	Default	Escalate when
Cost + owner	Tags Project=100Days Owner=personal Env=lab	Finance needs a policy to deny untagged RGs
Stop accidental delete	CanNotDelete lock on the RG	Prod: lock + PIM + deny-delete policy
Stop all writes	ReadOnly lock	You still need to rotate secrets — unlock first
Repeatable deploy	Bicep/ARM template later in the series	Click-ops that cannot be reviewed in Git
Provider missing	az provider register --namespace Microsoft.Web	Portal create blade hangs on Registering
Cleanup	Delete RG after removing locks	Lock left on → az group delete fails forever

Architecture C — pitfalls that make delete hang or cost unowned

Pitfall	What you see	Fix
Lock then delete	az group delete: Cannot delete resource group ... because of lock	az lock delete --name cannot-delete -g rg-day03-lab
Tags on resource not RG	Cost analysis cannot group by Project	Tag the RG AND resources; inherit is not magic
CanNotDelete on a VM NIC	RG delete still blocked by nested lock	az lock list -g ... -o table; delete each
ReadOnly during lab	az group update --tags fails	Use CanNotDelete for this lab, not ReadOnly
Provider not registered	Namespace Microsoft.Web not registered	az provider register --namespace Microsoft.Web
Leaving the lock in prod-think	Empty lab RG cannot be deleted next week	Remove lock, then --yes --no-wait delete

One-liner to remember

Tags are sticky notes for Finance. Locks are duct tape so nobody deletes the lab by accident.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription only. Create, lock, fail-delete, unlock, delete. Leave no billed resources.

- 1 `az group create -n rg-day03-lab -l centralindia --tags Project=100Days Owner=personal Env=lab`
- 2 `portal.azure.com` → `rg-day03-lab` → Tags. Confirm the three tags. Cost Management → Cost analysis → Group by Tag: Project.
- 3 `az lock create --name cannot-delete --lock-type CanNotDelete --resource-group rg-day03-lab`
- 4 `az group delete -n rg-day03-lab --yes` (expect failure while locked). Read the error; do not force anything else.
- 5 `az lock list --resource-group rg-day03-lab -o table` → `az lock delete --name cannot-delete --resource-group rg-day03-lab`
- 6 `az provider list --query [].namespace -o tsv | more` (Windows: `findstr Microsoft`). Note `Microsoft.Resources / Microsoft.Web`.
- 7 `az group delete -n rg-day03-lab --yes --no-wait`. Confirm `az group show -n rg-day03-lab` fails.

Tag the RG, lock it, prove delete fails, then unlock and destroy

```
az group create -n rg-day03-lab -l centralindia --tags Project=100Days Owner=personal Env=lab
az lock create --name cannot-delete --lock-type CanNotDelete --resource-group rg-day03-lab
az lock list --resource-group rg-day03-lab --output table
# Expect failure while locked:
# az group delete -n rg-day03-lab --yes
az lock delete --name cannot-delete --resource-group rg-day03-lab
az group delete -n rg-day03-lab --yes --no-wait
```

Done checklist

- Can explain ARM as the control plane behind Portal and CLI
- Applied Project/Owner/Env tags on `rg-day03-lab`
- Proved `CanNotDelete` blocks `az group delete`
- Removed the lock and deleted the RG

Tomorrow — Day 4

DevOps principles & culture